# ECS-PEC25/PEC33

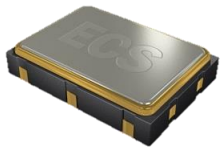
SMD PECL Oscillator

ECS-PEC25 (2.5V) and ECS-PEC33 (3.3V) miniature SMD PECL oscillators. Ideal for low jitter applications.

Request a Sample

## OPERATING CONDITIONS / ELECTRICAL CHARACTERISTICS

**ECS-PEC25/PEC33**



- Low Voltage PECL
- 7 x 5 mm Footprint
- Low Jitter
- PbFree/RoHS Compliant

| Parameters | Conditions | ECS-PEC25 (+2.5V) | | | ECS-PEC33 (+3.3V) | | | Units |
|---|---|---|---|---|---|---|---|---|
| | | MIN | TYP | MAX | MIN | TYP | MAX | |
| **Frequency Range** | | 40.0 | | 300.0 | 40.0 | | 300.0 | MHz |
| **Operating Temperature** | Standard | 0 | | +70 | 0 | | +70 | °C |
| | Extended (N Option) | -40 | | +85 | -40 | | +85 | °C |
| **Storage Temperature** | | -50 | | +125 | -50 | | +125 | °C |
| **Supply Voltage** | VDD | +2.375 | +2.5 | +2.625 | +3.135 | +3.3 | +3.465 | VDC |
| **Frequency Stability*** | Option A | | | ±100 | | | ±100 | PPM |
| | Option B | | | ±50 | | | ±50 | PPM |
| | Option C | | | ±25 | | | ±25 | PPM |
| **Input Current** | Pin 1 Open or VIH | | | 90 | | | 90 | mA |
| **Stand-by Current** | Pin 1 = VIL | | | 30 | | | 30 | µA |
| **Output Symmetry** | @ 50% VDD Level | | | 40/60 | | | 45/55 | % |
| **Rise and Fall Times** | 20% VDD to 80% Level | | | 1 | | | 1 | ns |
| **"0" Level** | VOL | | | +1.195 | | | +1.745 | VDC |
| **"1" Level** | VOH | +1.415 | | | +2.215 | | | VDC |
| **Output Load** | 50Ω into VDD -2V | | | | | | | |
| **Disable Delay Time** | | | | 200 | | | 200 | ns |
| **Enable/Startup Time** | | | | 10 | | | 10 | ms |
| **RMS Jitter** | 12 KHz to 20 MHz band | | | 1 | | | 1 | ps |
| **Aging** | | | | ±5 | | | ±5 | PPM |

## Part Numbering Guide: Example ECS-PEC33-1000-B-N-TR

| ECS - | Series - | Frequency Abbreviations - | Stability - | Temperature - | Packaging |
|---|---|---|---|---|---|
| ECS | PEC25 = +2.5V<br>PEC33 = +3.3V | 1000 = 100 MHz | A = ±100 ppm<br>B = ±50 ppm<br>C = ±25 ppm | Blank = -0 ~ 70°C<br>M = -20 ~ +70°C<br>N = -40 ~ +85°C | TR = Tape & Reel<br>500/Reel |

## Package Dimensions (mm)

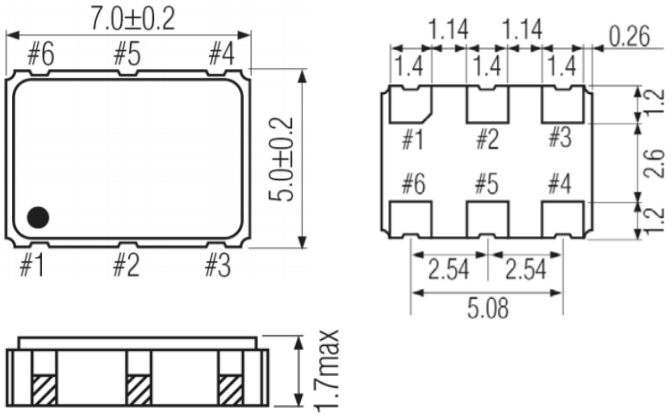



***Figure 1)*** *Top, Side, and Bottom views*

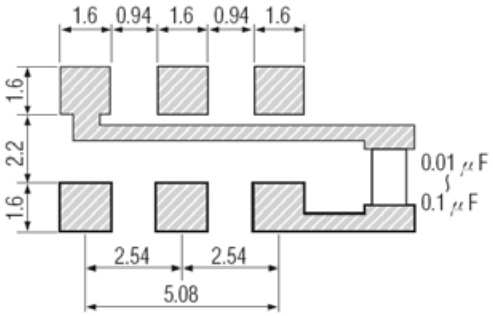



***Figure 2)*** *Land Pattern*

| Pin Connections | |
|---|---|
| #1 | Tri-State |
| #2 | N.C. |
| #3 | Ground |
| #4 | Output |
| #5 | C-Output |
| #6 | VDD |

| Tri-State Control Voltage | |
|---|---|
| Pad 1 | Pad 4 & 5 |
| Open | Oscillation |
| VIH 70% VDD Min | Oscillation |
| VIL 30% VDD Max | No Oscillation |

*Note: Internal crystal oscillation to be halted (Pin1 = VIL)*

### Frequency Abbreviations

| FREQUENCY MHz | CODE |
|---|---|
| 100.000 | 1000 |
| 106.250 | 1062.5 |
| 125.000 | 1250 |
| 155.520 | 1555.2 |
| 156.250 | 1562.5 |

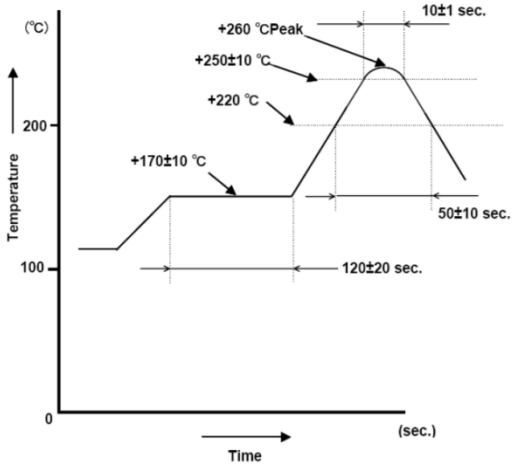
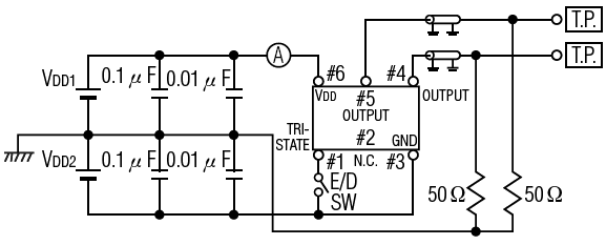


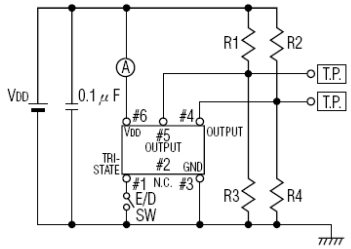

***Figure 3)*** *Suggested Reflow Profile*




Termination : 50Ω impedance matching

| VDD | VDD1 | VDD2 |
|---|---|---|
| +3.3V | +2.0V | -1.3V |
| +2.5V | +2.0V | -0.5V |

***Figure 4)*** *Test Circuit 1*




Termination : high impedance probes

| VDD | R1 | R2 | R3 | R4 |
|---|---|---|---|---|
| +3.3V | 130Ω | 130Ω | 82Ω | 82Ω |
| +2.5V | 270Ω | 270Ω | 62Ω | 62Ω |

Note : R3 & R4 to change for the use of low impedance probes

***Figure 5)*** *Test Circuit 2*

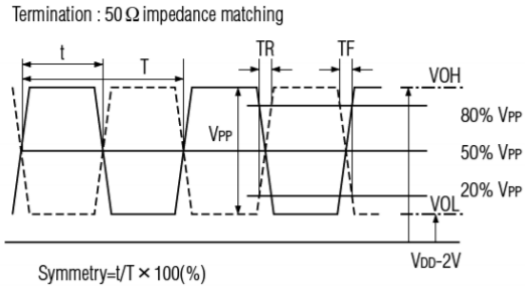



***Figure 6)*** *Output Waveform 1*

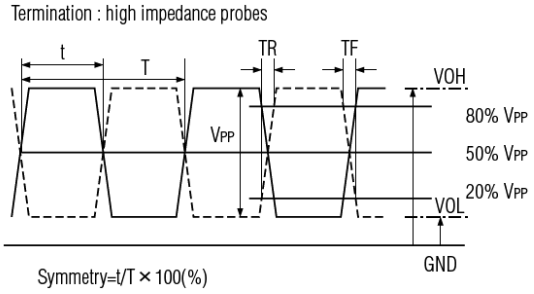
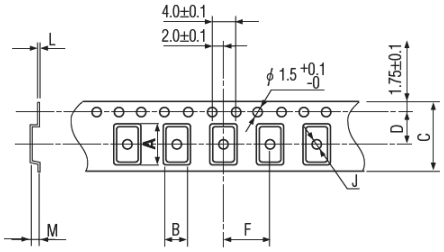



***Figure 7)*** *Output Waveform 2*

## Tape Dimensions (mm)




| A | B | C | D | F | J | L | M | Reel Dia. | Qty/Reel |
|---|---|---|---|---|---|---|---|---|---|
| 7.5 | 5.5 | 16.0 | 7.5 | 8.0 | 2.0 | 0.3 | 2.2 | 245 | 500pcs |

***Figure 8)*** *Pocket Tape Dimensions*

| Package Data | |
|---|---|
| **Item** | **Description** |
| Lid | Metal |
| Base | Ceramic |
| Sealing | Seam |
| Terminal | Tungsten (Metalized) |
| Plating | Gold/Nickel (Surface)/(Under) |
| RoHS | Compliant (PbFree) |